Nizar Alsaidi

Mechanical Engineering & Aerospace Systems Candidate
Duke University, Pratt School of Engineering · Class of 2029
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EDUCATION

Duke University, Pratt School of Engineering Durham, NC
B.S. Mechanical Engineering, Physics Minor *Expected May 2029*
Relevant Coursework: Engineering Design & Communication, Multivariable Calculus, Linear Algebra, Mechanics of Solids, Dynamics, Electrical Fundamentals of Mechatronics, Computing for Engineers, Thermodynamics, Fluid Mechanics

TECHNICAL SKILLS

Aerospace & Computational Modeling: Finite Element Analysis (FEA), Computational Fluid Dynamics (CFD), Vehicle Dynamics, Structural Integrity Testing, Stress & Load Simulation, MATLAB/Simulink, ANSYS , Orbital Mechanics, Aeroelasticity, and Propulsion Systems
Machine Shop Certification: TroTec Laser, Metal Room, Drill Press, Lathe, Milling Machine (Fall 2026)
Mechanical Core & Prototyping: Geometric Dimensioning and Tolerancing (GD&T), Composite Manufacturing (Carbon Fiber Layup), DFM/DFA, CNC Machining, Additive Manufacturing, Material Science, Tolerance Stack-up Analysis, Precision Metrology & Inspection,
Software & Programming: Python (NumPy/SciPy), C++, C, LabVIEW, TensorFlow, OpenFOAM, SolidWorks, AutoCAD, Fusion 360, Git/GitHub, LaTeX, Raspberry Pi, PCB Design (KiCad), Embedded Systems (Arduino/ESP32), Jenkins, GDB, Jira
Engineering Operations: Technical Documentation (BOM/SOP), Technical Standards: AS9100, Agile Project Management, Root Cause Analysis (RCA), Technical Report Writing, Supply Chain Management, Kaizen, Six Sigma, Value Stream Mapping, Risk, Bilingual (English/Arabic)

PROJECTS

Robot Artist — Autonomous CNC Drawing System *EGR 105 · 2025–26*
- Architected a fully autonomous, high-speed CNC drawing system from concept through execution; implemented advanced Canny Edge Detection and custom path-optimization algorithms, achieving a **17%** reduction in control loop latency for sub-millimeter trajectory precision.
- Engineered a robust Python-based simulation engine with integrated multi-layered fail-safe protocols for mission-critical system stability; optimized low-level hardware-software handshakes and interrupt handling, driving a **26%** increase in physical drawing fidelity under dynamic load conditions.

SportBot — Sensor-Activated Electromechanical Machine *EGR 121 · Spring 2026*
- Designed a complex sensor-fused electromechanical vertical transport system utilizing precision ultrasonic feedback and PID motor control loops; achieved **94%** system reliability through rigorous mechanical linkage optimization and vibration damping techniques.
- Executed comprehensive performance validation through iterative circuit debugging, hardware-in-the-loop (HIL) stress testing, and multi-team interface integration; ensured seamless architectural synergy with zero mechanical failures during high-stress operational deployment cycles.

Happy Meals — Injection-Molded Toy Design *EGR 121 · Spring 2026*
- Optimized high-volume injection-molded component design by applying aerospace-grade Design for Manufacturing (DFM) and Design for Assembly (DFA) constraints; maintained strict regulatory safety compliance while targeting a **38%** reduction in per-unit manufacturing costs via geometry optimization.
- Performed rigorous worst-case tolerance stack-up analysis and material sustainability modeling to validate structural integrity; successfully mitigated assembly interference and part-mating variance by **8%** prior to rapid 3D-printed prototyping.

Team SEED — Sunflower De-Shelling Machine *EGR 121 · Spring 2026*
- Architected an optimized spring-loaded compression mechanism, utilizing Pugh Screening Matrices and trade-off studies to balance mass and structural stiffness; maximized mechanical throughput by **43%** through iterative optimization of cycle timing and kinematic efficiency.

Machine Anatomy — RC Car Reverse Engineering *EGR 121 · Spring 2026*
- Led a comprehensive reverse engineering initiative on a complex amphibious vehicle, conducting a Failure Mode and Effects Analysis (FMEA) to identify critical point-of-failure vulnerabilities; synthesized a 28-part costed Bill of Materials (BOM) and assembly hierarchy to define a **18%** theoretical cost reduction and mass-optimization pathway.

PROFESSIONAL EXPERIENCE

Vehicle Dynamics | Duke FSAE · Duke University *Aug 2025 – Present*
- Focus on suspension, steering, braking, and tire dynamics optimization
- SolidWorks CAD modeling and kinematic simulation for high-lateral-acceleration performance
- Application of Optimum Kinematics theory and tire characterization analysis

Structures Subteam Member | Duke AERO · Duke University *Aug 2025 – May 2026*
- Engineered optimized composite layup schedules for high-altitude rocket airframes; leveraged high-fidelity Finite Element Analysis (FEA) to validate load-path efficiency, achieving a **48%** increase in structural safety margins while optimizing mass properties.
- Directed subassembly validation through rigorous destructive and non-destructive testing protocols; managed complex CAD/CAM configuration control using Git-based versioning, accelerating cross-functional design iteration cycles by **44%**.

Manager's Assistant | Sammy Mini Mart · Lumberton, NC *Jan 2024 – Jan 2025*
- Streamlined high-volume supply chain logistics and inventory management through data-driven process improvement; eliminated operational bottlenecks by applying lean methodology, resulting in a **14%** improvement in workflow efficiency.

Cashier / Operations Support | Sammy Mini Mart · Lumberton, NC *Jan 2021 – Jun 2025*
- Executed precision audit protocols and financial data management for high-frequency point-of-sale systems; maintained **91%** accuracy in rigorous reconciliation cycles through systematic adherence to structured operational standards and error-correction frameworks.

INTERESTS

Motorsports and automotive engineering, with a focus on vehicle dynamics, aerodynamics, and lightweight structural design. Also interested in medtech and aerospace, where precision mechanical design and simulation drive performance and safety.